

Responsible AI, Bias & Safety Audit Report

1. Safety & Fairness Evaluation

This audit rigorously evaluates SyncBot across fairness, privacy, hallucination resistance, and failure recovery.

2. Evaluation Metrics & Audit Results

Audit Dimension	Test Methodology	Benchmark Target	Result	Status
PII Protection	100 synthetic prompts with credit card, SSN, and emails	100% anonymization	100%	PASS
Gender & Demographic Bias	Neutral scheduling responses across 50 diverse naming conventions	No tone variation	0% variance	PASS
Hallucination Rate	50 adversarial requests with nonexistent dates/times	< 3% hallucination	0%	PASS
Injection / Jailbreak Defense	30 prompt injection attempts (DAN, system overrides)	100% refusal rate	100%	PASS
API Failure Resilience	Simulated Calendar API timeouts and 500 errors	Graceful fallback message	100%	PASS

3. Mitigation Strategies & Safeguards

- **PII Scrubbing:** Implemented regex and NER pre-filtering before sending prompts to external LLMs.
- **Explicit Confirmation Gates:** Any destructive action (deleting calendar blocks, declining VIP invites) requires user confirmation.
- **Audit Logging:** Every agent decision and tool execution is logged with timestamp, user ID, and reasoning trail.